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SEMINAR PAPER

# THE IMPACT OF GLOBALIZATION ON THE HUMAN DEVELOPMENT INDEX IN SOUTH AND SOUTHEAST ASIAN COUNTRIES

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# 1. ABSTRACT

This study aims to investigate the effect of globalization on human development with a panel of data from 1995 to 2015 for some South and South-East Asian countries. To correct the unobserved heterogeneity that characterizes the panel dataset, a Two-Ways Fixed Effects model was executed. However, after detecting the non-stationarity of the variables, a robust OLS regression with first-differences variables seems to correct the issue, providing results generally in line with the expectations. I ran three different regressions. One uses the first-differenced overall KOF Globalization Index as the main independent variable of interest; the second aims to analyze the impact of each of the three aspects of globalization: Economic, Social and Political globalization. The third specification includes two proxies for the economic side of globalization: the sum of imports and exports as per cent GDP, and the net inflows of Foreign Direct Investments as per cent GDP. Results reveal that the KOF Globalization Index had a significant effect on the HDI. More specifically, social globalization had a higher impact with respect to the other two sides. The openness to trade generated growth in the HDI; on the contrary, a negative effect has been encountered in the relationship between the FDIs and human development.

## 2. INTRODUCTION AND MOTIVATION

In the context of an open economy, more than 2/3 of the international trade occurs through Global Value Chains (ITC, 2019), and the production is made of a complex hubs network.

The Global Value Chain consists of splitting the production process of goods and services into a sequence of transnational phases, where each of them adds some value. The existence of these production networks has enhanced the economic connections between countries all over the world and this offered more opportunities to the developing countries in integrating into the global economy (ITC, 2019).

The concept of globalization included several definitions: firstly, it is a process by which the world economies are more integrated. With globalization, we refer also to an emerging “global culture” as well as financial globalization where financial flows and foreign direct investments are the main actors in international trade (Todaro & Smith, 2020)

The developing countries benefit from a globalized world through cultural, social, scientific and technological exchanges.

These flows encouraged the governments of several developing countries to enact development policies, and I would address my research question in this context, especially focusing on the South and South-East Asian countries.

## 3. THEORY

The Human Development Index is a geometric mean of normalized indexes used to compare the different statuses of socio-economic development (Todaro & Smith, 2020). Created by the United Nations Development Programme in 1990, it is a summary measure of achievement in 3 dimensions that characterize human development (United Nations Development Programme, 2022): a long and healthy life, measured with the life expectancy at birth, knowledge as the arithmetic mean of expected years of schooling combined with mean years of education, and a decent standard of living measured by the per capita gross domestic income in Purchasing Power Parity dollars. The

current scale ranks the countries according to the indexes between 0 (lowest) and 1 (highest).

Along with this technical definition, human development is depicted as “a process of enlarging people’s choices” (United Nations Development Programme, 1990). Here, the globalization process with the ongoing social, economic and cultural integration among nations and cultures can stimulate the improvement of the standards of living of the human society (Ulucak, et al., 2020).

Two different theories exist on the relationship between globalization and human development and one opposes the other. As described by Figueroa (2004), the first theory pioneered by Grennes in 2003 arrives from the neoliberal school that considers globalization as inflows and outflows of goods, services, assets, and capital technology that improves efficiency in the production process. Accordingly, it generates prosperity that can be spread over all the people, thus enhancing the population well-being (Grennes, 2003). In the debate on the consequences of globalization, some experts argued that expected costs associated with the phenomenon exceed its benefit, mainly concerning developing countries (Sapkota, 2011). Indeed, this “new world order”, illustrated by money flows and global powers, has induced a considerable accumulation of capital in the hands of a few private individuals at the expense of those who were already less privileged, undermining the population well-being (Figueroa, 2004). Due to the import of technology and machinery, the reduction in need of labor has created surplus labor issues. The movement of skilled labor from developing to developed countries caused brain drain in developing countries (Hasan & Waheed, 2020). As a result, it reduces human capital, limiting the capacity to innovate, and these factors have harmful effects on economic growth. Furthermore, recent literature (Wang, et al., 2018) concluded that economic growth, boosted by globalization from one side, guides to worsening the human development process. It may be mainly because of corruption, income inequality and inadequate allocation of the resources to strategical public sectors: education and health.

# 4. LITERATURE REVIEW

Some empirical studies on the relationship between globalization and human development have been presented.

Figueroa, 2004 examined globalization's impact on human development in 17 Latin American countries from 1995 to 2009. Using the Panel Corrected Standard Errors modelling, the author obtained a positive correlation between the overall KOF Globalization index and the Human Development Index. Nevertheless, the impact of the sub-indexes was more adverse. The economic globalization sub-index generated a negative effect; whereas both the political and social globalization sub-indexes had positive impacts on the HDI. The paper introduced, as control variables, the level of state fragility and delinquency in each country.

(Mohammad, et al., 2010) studied the consequence of globalization on HDI, with a case study concentrated on Pakistan. Running an OLS regression, they found that the coefficient of Foreign Direct Investments FDI shows a significant positive impact on the Human Development Index. That is justified since the FDI leads to an increase in employment opportunities, thus improving the standard of living.

They calculated that the Trade Openness, measured as the rate of change in the Balance of Trade, is significant and with a positive coefficient on the HDI.

# 5. EMPIRICAL PART

This study assembles a panel of 13 developing countries covering 10 years from 1995 to 2015. The original intention was to include the years 1990-2010 but, due to availability of data, I had to shift forward the time frame of 5 years, which is still consistent with my motivation, since 1995 is the year of the foundation of the World Trade Organization that signed a noteworthy formalization of the globalization process (WTO, 2022). The selected countries locate in the South and South-East of Asia: Bangladesh, Cambodia, India, Indonesia, Laos, Malaysia, Myanmar, Nepal, Pakistan, Philippines, Thailand, Sri Lanka and Vietnam.

Afghanistan, Brunei, Bhutan, Maldives and Timor Leste are removed from the study due to lack of data. Singapore is not part of the sample since Singapore is a high-income country and including it would collide with my intent.

About the dependent variable of interest, the Human Development Index for the panel of 13 Asia developing countries is extracted from the United Nations Development Programme (UNDP).

A further explanation is necessary for the globalization independent variable.

Globalization may be described using several different measurements. The KOF Index of Globalization, introduced by Dreher in 2006, measures globalization according to three dimensions: economic, social and political (Gygli, et al., 2019). Economic globalization accounts for the long-distance flows of goods, capital and services jointly with the data and perceptions that accompany market exchanges. Social globalization considers the spread of ideas, information, images and people, whereas political globalization represents the diffusion of government policies (Dreher, 2006).

Another measure, the openness to trade, measured as imports and exports as a percentage of the GDP, is frequently used as a proxy for globalization (Gygli, et al., 2019), together with the net inflows of Foreign Direct Investment as a share of GDP.

The KOF Globalization Index is pulled out from the KOF Swiss Economic Institute, while the openness to trade and the net inflows of FDI are from the World Bank Database. As control variables, I used the population growth rate, the birth rate and the death rate. The measures are taken from the United Nations, Department of Economic and Social Affairs, Population Division database.

State Fragility is specified when one or more of the three core functions of the state fail: state authority, state capacity and state legitimacy (Ziaja, et al., 2019) It can be considered dangerous for national, regional and international stability. Pieces of evidence showed that State Fragility is lower in countries with higher human development (Figueroa, 2004). I added State Fragility Index as a control

variable, taking the data from the Integrated Network for Societal Conflict Research INSCR Data Page[1]

Table 1: Summary Statistics

| Variable | N | Mean | Std. Dev. | Min | Pctl. 25 | Pctl. 75 | Max |
|---|---|---|---|---|---|---|---|
| Human Development Index | 273 | 580.623 | 105.714 | 380 | 493 | 669 | 796 |
| KOF Glob. Index | 273 | 51.705 | 13.172 | 22.771 | 40.962 | 61.508 | 80.935 |
| Economic KOF Glob. Index | 273 | 48.424 | 14.791 | 19.707 | 36.573 | 60.647 | 77.122 |
| Social KOF Glob. Index | 273 | 40.131 | 16.152 | 12.462 | 26.878 | 50.491 | 81.252 |
| Political KOF Glob. Index | 273 | 66.412 | 18.226 | 22.461 | 54.784 | 81.452 | 92.732 |
| Net Inflows FDI as percent GDP | 269 | 2.675 | 2.648 | -2.757 | 0.843 | 3.668 | 14.146 |
| Trade Openness | 268 | 77.694 | 48.949 | 0.167 | 42.216 | 111.465 | 220.407 |
| State Fragility Index | 273 | 12.725 | 3.807 | 5 | 10 | 16 | 20 |
| Population Growth Rate | 273 | 1.444 | 0.612 | -0.231 | 1.011 | 1.854 | 3.123 |
| Birth Rate | 273 | 23.433 | 5.952 | 10.836 | 18.576 | 27.493 | 38.417 |
| Death Rate | 273 | 7.169 | 1.584 | 4.499 | 5.867 | 8.181 | 11.904 |

Before initiating the analysis, I reviewed the distributions of the variables. Most of them are almost normally distributed. However, the presence of particularly skewed regressors advised transforming the variables by taking the logarithm or the squared root. Despite this, the transformation process made them even more skewed, so I will preserve the original variable types, being aware of this limitation.

This empirical analysis will comprehend three different specifications. The first specification has the overall KOF Index as unique globalization independent variable:

$$HDI_{it} = \beta_0 + \beta_1 KOF_{it} + \beta_2 StateFragility_{it} + \beta_3 PopGrowth_{it} + \beta_4 BirthRate_{it} + \beta_5 DeathRate_{it} + e_{it}$$

In the second specification, all three sub-indexes will be taken into account as regressors:

$$HDI_{it} = \beta_0 + \beta_1 KOFE_{it} + \beta_2 KOFS_{it} + \beta_3 KOFP_{it} + \beta_4 StateFragility_{it} + \beta_5 PopGrowth_{it} + \beta_6 BirthRate_{it} + \beta_6 DeathRate_{it} + e_{it}$$

[1] Open-Access source http://www.systemicpeace.org/inscrdata.html

The revisited version of the KOF Globalization Index distinguishes globalization between de jure and de facto. The former regards policies and conditions that ease cross-border activities, whereas the latter estimates actual international flows and activities (Gygli, et al., 2019). The overall KOF Globalization Index merges de facto and de jure globalization; meanwhile, for the three sub-indexes, I focused the attention on de facto globalization.

A third specification estimates the effect of globalization using trade openness and the FDI measure as proxies.

$$HDI_{it} = \beta_0 + \beta_1 TradeOpenness_{it} + \beta_2 FDI_{it} + \beta_3 StateFragility_{it} + \beta_4 PopGrowth_{it} + \beta_5 BirthRate_{it} + \beta_6 DeathRate_{it} + e_{it}$$

There is a drawback regarding Myanmar data: for both measures, openness to trade and net FDI inflows, no database reported years 1995 to 1999, and 1995 to 1998, respectively. Since these missing data are caught at the beginning of the dataset, I will proceed further, including the country in the sample.

Table 2 Correlation Matrix

| | HDI | KOF | KOFE | KOFS | KOFP | FDI | Openness | Fragility | PopGrowth | BirthRate | DeathRate |
|---|---|---|---|---|---|---|---|---|---|---|---|
| HDI | 1 | 0.858 | 0.703 | 0.881 | 0.459 | 0.059 | 0.531 | -0.780 | -0.502 | -0.788 | -0.749 |
| KOF | 0.858 | 1 | 0.760 | 0.919 | 0.698 | 0.054 | 0.509 | -0.750 | -0.226 | -0.611 | -0.685 |
| KOFE | 0.703 | 0.760 | 1 | 0.698 | 0.136 | 0.423 | 0.804 | -0.731 | -0.110 | -0.490 | -0.530 |
| KOFS | 0.881 | 0.919 | 0.698 | 1 | 0.490 | 0.122 | 0.572 | -0.775 | -0.332 | -0.644 | -0.704 |
| KOFP | 0.459 | 0.698 | 0.136 | 0.490 | 1 | -0.387 | -0.134 | -0.289 | -0.091 | -0.320 | -0.394 |
| FDI | 0.059 | 0.054 | 0.423 | 0.122 | -0.387 | 1 | 0.521 | -0.338 | 0.038 | -0.111 | -0.124 |
| Openness | 0.531 | 0.509 | 0.804 | 0.572 | -0.134 | 0.521 | 1 | -0.721 | -0.063 | -0.446 | -0.487 |
| Fragility | -0.780 | -0.750 | -0.731 | -0.775 | -0.289 | -0.338 | -0.721 | 1 | 0.398 | 0.788 | 0.654 |
| PopGrowth | -0.502 | -0.226 | -0.110 | -0.332 | -0.091 | 0.038 | -0.063 | 0.398 | 1 | 0.787 | 0.303 |
| BirthRate | -0.788 | -0.611 | -0.490 | -0.644 | -0.320 | -0.111 | -0.446 | 0.788 | 0.787 | 1 | 0.595 |
| DeathRate | -0.749 | -0.685 | -0.530 | -0.704 | -0.394 | -0.124 | -0.487 | 0.654 | 0.303 | 0.595 | 1 |

The table above shows the correlation matrix of the no string variables and as expected, the overall KOF Globalization Index highly correlates with the three KOF components. Therefore, running two separate regressions would avoid the imperfect multicollinearity issue that may arise.

I started the empirical analysis on R declaring that the dataset has a panel setting. R provides additional information that the panel is balanced, therefore we have the same time periods *T* for each observation unit *i* (Wooldridge, 2013), with 13 countries and 21 years.

The panel data techniques can handle unobserved heterogeneity, which might bias the estimation. Consequently, the econometrics theory would suggest running three different kinds of estimation models (Pooled OLS, Fixed Effect (with the following specifications One-Way or Two-Way depending on the presence of individual and time effects) and the Random Effect model.

After running the Pooled OLS model and the One-Way Fixed Effects model, I tested the presence of individual effects with the F-test. The result suggested the presence of unobserved heterogeneity and thus preferring the Fixed Effect model over the Pooled OLS. I checked for the significance of the fixed effects, and they are all highly significant at a 1% level. Then, I tested the absence of time effects using the F-test for the Two-Ways Fixed Effect, and as expected, since the test strongly rejects the null, the One-Way Fixed Effects model cannot be used. I also ran the Breusch Pegan Lagrange Multiplier Test to confirm the Two-Way Fixed Effect model over the Pooled OLS model.

I implemented the Random Effects model. I executed the Hausman Test to check whether the idiosyncratic errors are correlated or not with the regressors. Rejecting the Hausman Test, the time-invariant fixed effects correlate with the regressors: the Random Effects model cannot be used in this context, since the assumption of $cov(\alpha_i, x_{it}) = 0$ is violated.

The Two-Way Fixed Effect seems to be the most suitable model to implement in all three specifications.

I proceeded to test some assumptions: no serial correlation in idiosyncratic errors, no cross-sectional dependence, homoscedasticity.

The null hypothesis of the Breusch Pegan test is that the error variances are all equal.

Since there is evidence for heteroskedasticity, a robust covariance matrix is advised.

I conducted the Pesaran cross-sectional dependence test to check for spatial dependence in the model. Rejecting the null hypothesis, the model violates the cross-section independence assumption.

After running the Breusch-Godfrey/Wooldridge test for serial correlation in panel models, I strongly rejected the null of the absence of serial correlation in

idiosyncratic errors. Since the evidence of temporal dependence in the model is quite strong, I inspected whether the variables are stationary or non-stationary. To detect whether the variables are stationary or not stationary, I implemented the cross-sectionally augmented panel unit root test (CIPS test) proposed by (Pesaran, 2007). This “second-generation panel unit roots test” considers the correlation between cross-sections. The outcomes recommend using the First Difference approach, given the non-stationarity of the variables.

With first differencing, the time-invariant fixed effects and the constant will disappear.

To execute a first-difference of the Two-Way Fixed Effect model, I first-differenced all the variables, created the time dummies, and I first-differenced the time dummies.

I ran a robust OLS model using the first-differenced variables and including the first-differenced time dummies to control for country trends.

The following table reports the outcomes of all three robust OLS multivariate specifications with first-differenced variables, adding the first-differenced year dummies as control variables.

In the first column, the coefficients from the regression predicting the impact of the overall KOF Globalization Index on the Human Development Index indicates that the latter will increase by 4.494 if the KOF Globalization Index changes by 1 unit. This result is highly significant. The regression in the second column reflects the influence of the three components of the KOF Globalization Index. The results demonstrate that growth in the Social KOF Globalization Index would positively impact the Human Development Index more than in the case of an increase in the Economic and Political KOF Globalization Index, precisely by 2.8 units against an increase by 0.7 units or 1.07 respectively. The third specification focuses on the economic side of the globalization process, i.e. Net Inflows of Foreign Direct Investment as a percentage of the GDP and Trade Openness as a percentage of the GDP.

Raising the Trade Openness variable had a positive and significant effect at 10% level on the dependent variable of interest, increasing by 0.34 units. Contrariwise,

the net inflows FDI present a negative and highly significant effect on the Human Development Index, decreasing by 2.2 units for every one-unit increase in the independent variable. In all three specifications, the control variables are not statistically significant, except the State Fragility Index, since it is significant in the first and third specification at 10% and 5% level of significance, respectively.

Expectedly, as the country works on enforcing the institutions and democracy and on diminishing the violent environment, the influence on human development is positive. In these regressions, the Year Dummies are included. For the first specification, only the year dummies from dY2005 to dY2010 are significant.

While no one is significant in the second regression, in contrast, the third specification delivers highly statistically significant coefficients for almost all the year dummies counted in the model.

The adjusted R-squared is very high in all three specifications: it suggests that a good proportion of the variation in the Human Development Index is explained by the independent variables.

The probability of F-statistic shows that the model is overall statistically significant.

For this empirical analysis, I added a robustness check (Table 3) to explore how the regression coefficient estimates behave excluding the year dummies.

No significant difference is witnessed in the first specification. In the second specification, the Economic KOF Globalization Index delivers nonsignificant results, while the magnitude of the Social KOF Globalization Index coefficient is still larger than the one in the case of the Political KOF Globalization Index, as detected in the previous analysis. The third specification, on the other hand, presents adverse results: both the coefficients turned to be insignificant. This means that the openness to trade and the net inflows of FDI have no significant effect on human development, when year dummies are not considered as control variables in the model.

Table 3 Robust OLS regression with first-differenced variables, Year Dummies included

| | *Dependent variable:* | | |
|---|---|---|---|
| | Human Development Index | | |
| | (1) | (2) | (3) |
| KOF Globalization Index | 4.494*** | | |
| | (0.650) | | |
| Economic KOF Globalization Index | | 0.699** | |
| | | (0.331) | |
| Social KOF Globalization Index | | 2.769*** | |
| | | (0.614) | |
| Political KOF Globalization Index | | 1.072*** | |
| | | (0.409) | |
| Net Inflows FDI as percent GDP | | | -2.231** |
| | | | (1.046) |
| Trade Openness as percent GDP | | | 0.342* |
| | | | (0.196) |
| State Fragility Index | -2.271* | -1.741 | -2.621** |
| | (1.247) | (1.115) | (1.315) |
| Population Growth Rate | -17.998 | -13.316 | 4.859 |
| | (22.284) | (22.684) | (19.565) |
| Birth Rate | -1.924 | -1.875 | -5.891 |
| | (3.555) | (3.599) | (3.613) |
| Death Rate | -5.880 | -7.273 | -11.310 |
| | (4.571) | (4.429) | (8.406) |
| (Year Dummies not shown) | | | |
| Observations | 272 | 272 | 266 |
| $R^2$ | 0.903 | 0.910 | 0.827 |
| Adjusted $R^2$ | 0.894 | 0.900 | 0.808 |
| Residual Std. Error | 13.212 | 12.809 | 14.011 |
| F Statistic | 92.306*** | 91.591*** | 44.188*** |

*Note:* *p<0.1; **p<0.05; ***p<0.01
Robust Standard Errors in parentheses

Table 4: Robust OLS regression with first-differenced variables
Year Dummies excluded

| | *Dependent variable:* | | |
|---|---|---|---|
| | Human Development Index | | |
| | (1) | (2) | (3) |
| KOF Gloablization Index | 4.580*** | | |
| | (0.832) | | |
| Economic KOF Globalization Index | | 0.638 | |
| | | (0.465) | |
| Social KOF GLobalization Index | | 2.613*** | |
| | | (0.770) | |
| Political KOF Globalization Index | | 1.044*** | |
| | | (0.353) | |
| Net Inflows FDI as percent GDP | | | -0.994 |
| | | | (0.675) |
| Trade Openness as percent GDP | | | 0.166 |
| | | | (0.194) |
| State Fragility Index | -2.229** | -1.699 | -4.087*** |
| | (1.099) | (1.186) | (1.517) |
| Population Growth Rate | -18.893 | -8.157 | -1.780 |
| | (25.858) | (21.958) | (24.363) |
| Birth Rate | -2.423 | -2.117 | -8.085** |
| | (3.312) | (3.201) | (3.259) |
| Death Rate | -5.608 | -7.606* | -9.828 |
| | (4.086) | (4.367) | (9.387) |
| Observations | 272 | 272 | 266 |
| $R^2$ | 0.895 | 0.906 | 0.784 |
| Adjusted $R^2$ | 0.893 | 0.904 | 0.779 |
| Residual Std. Error | 13.224 | 12.577 | 15.043 |
| F Statistic | 456.602*** | 364.881*** | 157.482*** |

*Note:* *p<0.1; **p<0.05; ***p<0.01
Robust Standard Errors in parentheses

We can deduce that, at least for the first two specifications, the coefficients of the main independent variables of interest seems to be reasonable and robust. Whereas to obtain significant outcomes in the third specification, counting year dummies is crucial. When all the variables are non-stationary, the panel

cointegration methodology can also be implemented as alternative technique. However, a detailed and thorough implementation of this method is beyond the scope of the course materials. Despite this, I have checked external sources, yet I have not been able to find a solid guide for my analysis. Considering these limitations, I would stick to my analysis using a robust OLS regression with first-differenced variables.

## 6. CONCLUSION

This study produces interesting results, showing that globalization plays a positive role in promoting human development. However, the magnitude of the effect is diverse, although still positive, depending on which type of globalization we take into account.

In the Social KOF Globalization Index, interpersonal, informational and cultural globalization are included; apparently, the HDI components life expectancy and education factor are more sensitive to changes in the social and political side of globalization, rather than to economic globalization changes. This insight is justified by looking at the third regression, which considers only the effect of an increase in economic integration.

Not only the impact of openness to trade is tiny, but direct investments made by non-resident investors towards the developing country does not improve the population well-being.

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# 8. APPENDIX
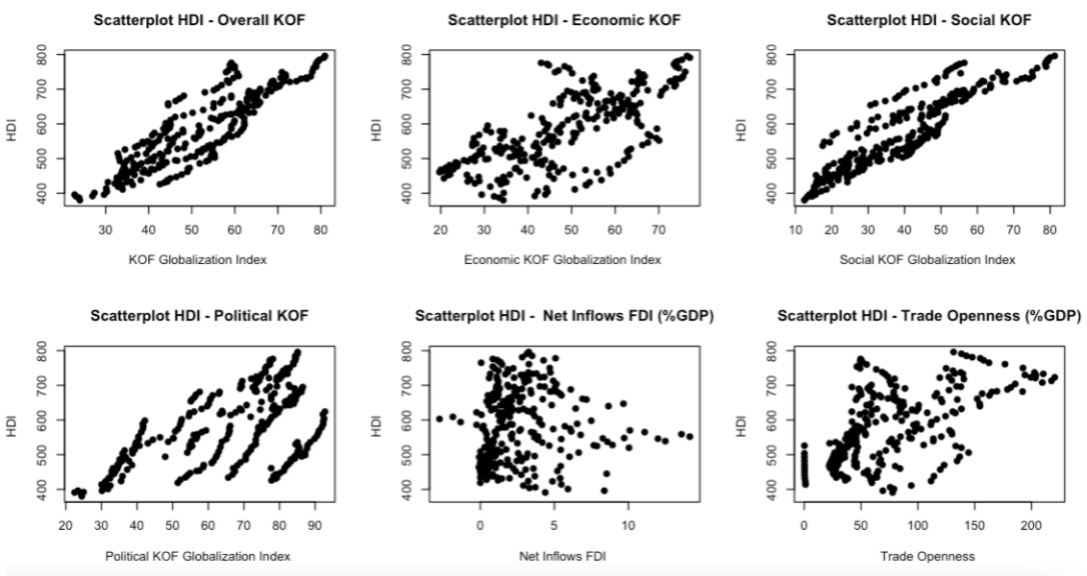
Table 5: Scatterplots - Human Development Index with the explanatory variables of interest